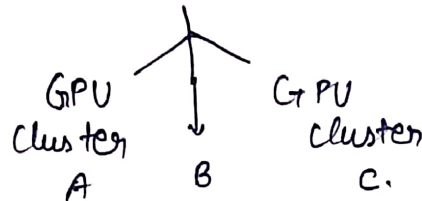


*What happens when you call any LLM API

① API Gateway.

Request hit → TLS Termination → Authentication - Rate limiter
→ API Key Valid
→ User Authorize.
→ Request format

② Load Balancer



@codingdidi

→ Geographic routing
→ least connection
→ Health checks.

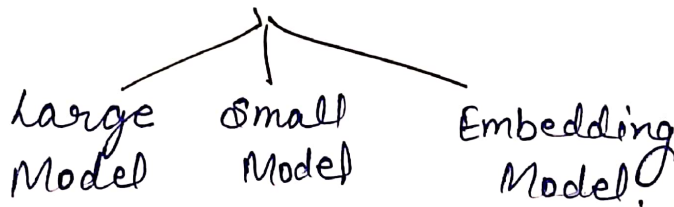
③ Tokenization

"Hello World" → [13359, 1917]

→ BPE
→ Sentence piece
→ WordPiece.

More token = More Cost

④ Model Router.



→ Model Version
→ Capacity-aware
→ Queue Management during peak load.

⑤ Inference Engine

→ Prefill Phase :- All token → process - Attention → KV Cache.

→ Decode Phase :- KV Cache + Previous token → Attention layer → FFN Layers

sample Next token ← softmax

→ Attention Mechanism



Hardware layers.

↳ processing in GPU.

⑥ Post Processing

Generated Tokens → Detokenisation → safety Classifier → Formatted Response.

⑦ Response & Billing.

→ JSON Response → streaming chunk → Back through Load Balancer.

↓

TLS
code.

@wdjddi

⑧ Logging & Observability

↳ Every call logged: latency, token, model, finish reason, safety flag